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Project: Online Ticket Booking Management System

Subject: Database Management system

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Project Report

1. Introduction

The Online Ticket Booking System is designed for managing bus ticket bookings for customers. It enables users to search for available buses, book seats, make payments, and manage ticket details. Bus operators and administrators can manage buses, routes, schedules, and payments efficiently through the system.

2. Database Structure

The database consists of multiple interrelated tables to manage data for customers, buses, routes, schedules, tickets, and payments. Here's a summary of each table:

- **Customers Table:** Contains customer information such as ID, name, email, phone number, and address.
- **Buses Table:** Stores bus details including bus number, type, capacity, and the operator's name.
- **Routes Table:** Details the different routes with origin, destination, distance, and duration.
- **Schedules Table:** Links buses to routes, and includes details like departure time, arrival time, and price.
- **Seats Table:** Manages seat availability for each schedule, including seat numbers and their status (available or booked).
- **Tickets Table:** Contains information about the tickets booked by customers, including the schedule, seat, and ticket price.
- **Payments Table:** Records payment transactions for the tickets, including the method, date, and amount.

Table 1: Customer Table

	⚡ CUSTOMER_ID	⚡ NAME	⚡ EMAIL	⚡ PHONE_NUMBER	ADDRESS	
1	1	Alice Smith	alice@example.com	1234567890	123 Main St	
2	2	Bob Johnson	bob@example.com	9876543210	456 Elm St	
3	3	Charlie Brown	charlie@example.com	5556667777	789 Pine St	
4	4	Diana Prince	diana@example.com	2223334444	101 Oak St	
5	5	Eve Adams	eve@example.com	1112223333	202 Maple St	
6	6	Frank Wright	frank@example.com	4445556666	303 Cedar St	
7	7	Grace Hall	grace@example.com	3334445555	404 Birch St	
8	8	Hank Miller	hank@example.com	7778889999	505 Spruce St	
9	9	Ivy Clark	ivy@example.com	8889990000	606 Cherry St	
10	10	Jack Davis	jack@example.com	9990001111	707 Ash St	

Table 2: Bus Table

	⚡ BUS_ID	⚡ BUS_NUMBER	⚡ BUS_TYPE	⚡ CAPACITY	⚡ OPERATOR_NAME	
1	1	AB123	AC Sleeper	40	Green Line Travels	
2	2	CD456	Non-AC Seater	50	Blue Horizon	
3	3	EF789	AC Seater	30	Silver Fleet	
4	4	GH101	Non-AC Sleeper	45	Red Star	
5	5	IJ112	AC Luxury	25	Gold Line	
6	6	AB123	AC Sleeper	40	Green Line Travels	
7	7	CD456	Non-AC Seater	50	Blue Horizon	
8	8	EF789	AC Seater	30	Silver Fleet	
9	9	GH101	Non-AC Sleeper	45	Red Star	
10	10	IJ112	AC Luxury	25	Gold Line	

Table 3: Routes Table

	⚡ ROUTE_ID	⚡ ORIGIN	⚡ DESTINATION	⚡ DISTANCE	⚡ DURATION	
1	1	City A	City B	200	180	
2	2	City B	City C	150	120	
3	3	City C	City D	300	240	
4	4	City D	City E	100	90	
5	5	City A	City E	400	360	
6	6	City A	City B	200	180	
7	7	City B	City C	150	120	
8	8	City C	City D	300	240	
9	9	City D	City E	100	90	
10	10	City A	City E	400	360	

Table 4: Schedule

	SCHEDULE_ID	BUS_ID	ROUTE_ID	DEPARTURE_TIME	ARRIVAL_TIME	PRICE
1	1	1	1	20-DEC-24 08.00.00.000000000 AM	20-DEC-24 11.00.00.000000000 AM	500
2	2	2	2	20-DEC-24 09.00.00.000000000 AM	20-DEC-24 11.00.00.000000000 AM	300
3	3	3	3	20-DEC-24 12.00.00.000000000 PM	20-DEC-24 04.00.00.000000000 PM	600
4	4	4	4	20-DEC-24 02.00.00.000000000 PM	20-DEC-24 03.30.00.000000000 PM	200
5	5	5	5	21-DEC-24 07.00.00.000000000 AM	21-DEC-24 01.00.00.000000000 PM	800
6	6	1	1	20-DEC-24 08.00.00.000000000 AM	20-DEC-24 11.00.00.000000000 AM	500
7	7	2	2	20-DEC-24 09.00.00.000000000 AM	20-DEC-24 11.00.00.000000000 AM	300
8	8	3	3	20-DEC-24 12.00.00.000000000 PM	20-DEC-24 04.00.00.000000000 PM	600
9	9	4	4	20-DEC-24 02.00.00.000000000 PM	20-DEC-24 03.30.00.000000000 PM	200
10	10	5	5	21-DEC-24 07.00.00.000000000 AM	21-DEC-24 01.00.00.000000000 PM	800

Table 5: Tickets Table

	TICKET_ID	CUSTOMER_ID	SCHEDULE_ID	SEAT_ID	BOOKING_DATE	TICKET_PRICE
1	1	1	1	1	18-DEC-24 02.47.57.000000000 PM	500
2	2	2	2	1	18-DEC-24 02.47.57.000000000 PM	500
3	3	3	3	2	18-DEC-24 02.47.57.000000000 PM	300
4	4	4	4	3	18-DEC-24 02.47.57.000000000 PM	600
5	5	5	5	4	18-DEC-24 02.47.57.000000000 PM	200
6	6	6	6	5	18-DEC-24 02.47.57.000000000 PM	800
7	7	7	7	1	18-DEC-24 02.47.57.000000000 PM	500
8	8	8	8	2	18-DEC-24 02.47.57.000000000 PM	300
9	9	9	9	3	18-DEC-24 02.47.57.000000000 PM	600
10	10	10	10	4	11 18-DEC-24 02.47.57.000000000 PM	200

Table 6: Seats Table

	SEAT_ID	SCHEDULE_ID	SEAT_NUMBER	IS_AVAILABLE
1	1	1	1 A1	Y
2	2	2	1 A2	Y
3	3	3	1 A3	Y
4	4	4	2 B1	Y
5	5	5	2 B2	Y
6	6	6	2 B3	Y
7	7	7	3 C1	Y
8	8	8	3 C2	Y
9	9	9	3 C3	Y
10	10	4	4 D1	Y

Table 7: Payments Table

	PAYMENT_ID	TICKET_ID	PAYMENT_DATE	PAYMENT_AMOUNT	PAYMENT_METHOD
1	1	1	18-DEC-24 02.49.09.000000000 PM	500	Credit Card
2	2	2	18-DEC-24 02.49.09.000000000 PM	500	Debit Card
3	3	3	18-DEC-24 02.49.09.000000000 PM	300	Net Banking
4	4	4	18-DEC-24 02.49.09.000000000 PM	600	UPI
5	5	5	18-DEC-24 02.49.09.000000000 PM	200	Cash
6	6	6	18-DEC-24 02.49.09.000000000 PM	800	Credit Card
7	7	7	18-DEC-24 02.49.09.000000000 PM	500	Debit Card
8	8	8	18-DEC-24 02.49.09.000000000 PM	300	Net Banking
9	9	9	18-DEC-24 02.49.09.000000000 PM	600	UPI
10	10	10	18-DEC-24 02.49.09.000000000 PM	200	Cash

3. SQL Queries

Various SQL queries were written to retrieve data from these tables, using different types of joins:

- **LEFT JOIN:** Retrieves customers and their tickets, even if they have not booked one.
- **RIGHT JOIN:** Retrieves tickets and their associated customer details, even if the ticket is unlinked.
- **FULL OUTER JOIN:** Shows all customers and tickets, even if some customers haven't booked a ticket or some tickets are unlinked.
- **INNER JOIN:** Retrieves customers who have booked tickets along with their ticket details.

Query 1: Left Join

	CUSTOMER_ID	NAME	TICKET_ID	BOOKING_DATE
1	1	Alice Smith	1	18-DEC-24 02.47.57.000000000 PM
2	2	Bob Johnson	2	18-DEC-24 02.47.57.000000000 PM
3	3	Charlie Brown	3	18-DEC-24 02.47.57.000000000 PM
4	4	Diana Prince	4	18-DEC-24 02.47.57.000000000 PM
5	5	Eve Adams	5	18-DEC-24 02.47.57.000000000 PM
6	6	Frank Wright	6	18-DEC-24 02.47.57.000000000 PM
7	7	Grace Hall	7	18-DEC-24 02.47.57.000000000 PM
8	8	Hank Miller	8	18-DEC-24 02.47.57.000000000 PM
9	9	Ivy Clark	9	18-DEC-24 02.47.57.000000000 PM
10	10	Jack Davis	10	18-DEC-24 02.47.57.000000000 PM

Query 2: Right Join

	TICKET_ID	BOOKING_DATE	NAME	EMAIL
63	63	18-DEC-24 03.43.01.0000000000 PM	Charlie Brown	charlie@example.com
64	64	18-DEC-24 03.43.01.0000000000 PM	Diana Prince	diana@example.com
65	65	18-DEC-24 03.43.01.0000000000 PM	Eve Adams	eve@example.com
66	66	18-DEC-24 03.43.01.0000000000 PM	Frank Wright	frank@example.com
67	67	18-DEC-24 03.43.01.0000000000 PM	Grace Hall	grace@example.com
68	68	18-DEC-24 03.43.01.0000000000 PM	Hank Miller	hank@example.com
69	69	18-DEC-24 03.43.01.0000000000 PM	Ivy Clark	ivy@example.com
70	70	18-DEC-24 03.43.02.0000000000 PM	Jack Davis	jack@example.com
71	(null)	(null)	Nathan Young	nathan@example.com
72	(null)	(null)	Olivia King	olivia@example.com

Query 3: Full outer Join

	CUSTOMER_ID	CUSTOMER_NAME	EMAIL	TICKET_ID	BOOKING_DATE	TICKET_PRICE
1	1	Alice Smith	alice@example.com	1	18-DEC-24 02.47.57.0000000000 PM	500
2	2	Bob Johnson	bob@example.com	2	18-DEC-24 02.47.57.0000000000 PM	500
3	3	Charlie Brown	charlie@example.com	3	18-DEC-24 02.47.57.0000000000 PM	300
4	4	Diana Prince	diana@example.com	4	18-DEC-24 02.47.57.0000000000 PM	600
5	5	Eve Adams	eve@example.com	5	18-DEC-24 02.47.57.0000000000 PM	200
6	6	Frank Wright	frank@example.com	6	18-DEC-24 02.47.57.0000000000 PM	800
7	7	Grace Hall	grace@example.com	7	18-DEC-24 02.47.57.0000000000 PM	500
8	8	Hank Miller	hank@example.com	8	18-DEC-24 02.47.57.0000000000 PM	300
9	9	Ivy Clark	ivy@example.com	9	18-DEC-24 02.47.57.0000000000 PM	600
10	10	Jack Davis	jack@example.com	10	18-DEC-24 02.47.57.0000000000 PM	200

Query 4: Inner Join

	CUSTOMER_ID	CUSTOMER_NAME	EMAIL	TICKET_ID	BOOKING_DATE	TICKET_PRICE
1	1	Alice Smith	alice@example.com	1	18-DEC-24 02.47.57.0000000000 PM	500
2	2	Bob Johnson	bob@example.com	2	18-DEC-24 02.47.57.0000000000 PM	500
3	3	Charlie Brown	charlie@example.com	3	18-DEC-24 02.47.57.0000000000 PM	300
4	4	Diana Prince	diana@example.com	4	18-DEC-24 02.47.57.0000000000 PM	600
5	5	Eve Adams	eve@example.com	5	18-DEC-24 02.47.57.0000000000 PM	200
6	6	Frank Wright	frank@example.com	6	18-DEC-24 02.47.57.0000000000 PM	800
7	7	Grace Hall	grace@example.com	7	18-DEC-24 02.47.57.0000000000 PM	500
8	8	Hank Miller	hank@example.com	8	18-DEC-24 02.47.57.0000000000 PM	300
9	9	Ivy Clark	ivy@example.com	9	18-DEC-24 02.47.57.0000000000 PM	600
10	10	Jack Davis	jack@example.com	10	18-DEC-24 02.47.57.0000000000 PM	200

Conclusion

The **Online Ticket Booking System** is a robust database management system designed to handle bus ticket reservations and associated processes efficiently. By integrating multiple tables, the

system provides a seamless flow of information between customers, buses, routes, schedules, seats, tickets, and payments.

Key features include:

- **Customer Management:** The system allows customers to book tickets, manage personal details, and track their booking history.
- **Bus and Route Management:** Operators can manage different types of buses, their capacities, and assigned routes, ensuring that the system can handle various travel options.
- **Ticket Booking and Payment:** Customers can book tickets for specific schedules, select available seats, and make payments through multiple methods. The system records payment transactions, ensuring smooth financial tracking.
- **Efficient Data Retrieval:** By utilizing SQL queries with different types of joins, the system ensures that all relevant data can be accessed efficiently, whether for customer queries or administrative tasks.